

```

1 D:\Software\Python\Python311\python.exe E:\PythonProjectsByYear\Year2025\MC-
  PIKAN_Official\experiments\Volterra1D_MCPi.py
2 The network model is PIKAN
3 Layers: ModuleList(
4   (0): ChebyKANLayer(in_features=1, out_features=10, degree=6)
5   (1): ChebyKANLayer(in_features=10, out_features=10, degree=6)
6   (2): ChebyKANLayer(in_features=10, out_features=1, degree=6)
7 )
8 Total trainable parameters: 840
9 Model on device: cuda:0
10 Start training...
11 Max Epochs: 2000, Learning Rate: 0.001, Epsilon: 1e-20, Optimizer: Adam
12 Training: 100%|██████████| 2000/2000 [00:09<00:00, 218.71epoch/s, Loss=1.
  93e-06, LR=1.0e-03, Best=1.93e-06]
13 Training finished. Best loss: 1.30e-06. Model saved to results/best_model.pth
14 Loss history saved to results/loss_history.npy
15 Training time: 9.947881698608398s
16 Relative loss: 0.0019332696683704853
17 L2 loss: 0.06112006679177284
18 Parameter containing:
19 tensor([[[[-0.0059, -0.0216,  0.2507,  0.0392,  0.0846, -0.1715, -0.1293],
20           [-0.0732, -0.0538,  0.1786,  0.1878,  0.0643, -0.0312, -0.0139],
21           [-0.0383,  0.2313,  0.0820, -0.0277, -0.1580,  0.0210, -0.3804],
22           [ 0.0027,  0.1309,  0.0266,  0.0007,  0.1037,  0.1388, -0.0971],
23           [ 0.1289,  0.1761,  0.0483,  0.2496,  0.1702,  0.2383,  0.0896],
24           [-0.1018, -0.0011,  0.0555,  0.2382, -0.0298, -0.0694,  0.0839],
25           [-0.3406,  0.1113, -0.1987, -0.1614, -0.1067,  0.0964,  0.0391],
26           [-0.1744,  0.1370, -0.1764,  0.0677, -0.2432,  0.2172,  0.0006],
27           [ 0.0254,  0.0269,  0.0890,  0.1621,  0.0598,  0.0007,  0.0993],
28           [-0.0484, -0.0352,  0.1199,  0.0510, -0.0056,  0.1068, -0.1307]]]],
29         device='cuda:0', requires_grad=True)
30 Parameter containing:
31 tensor([[[[-1.3832e-02,  4.9615e-03,  1.4566e-02,  2.0088e-03, -3.5993e-02,
32           -4.0575e-02,  1.3739e-02],
33           [ 1.5234e-02, -5.3473e-02,  1.3651e-02,  2.6205e-02, -6.2109e-03,
34           -2.8188e-02, -5.2224e-03],
35           [ 1.0845e-02,  1.3603e-02, -7.4279e-03, -1.4002e-03,  2.6115e-02,
36           -3.3222e-03,  1.8573e-03],
37           [ 4.4223e-02,  2.2690e-02, -3.8518e-02, -1.7684e-02,  2.4661e-02,
38           1.1411e-02, -2.0173e-02],
39           [-6.0943e-03, -1.7727e-02, -1.0509e-02, -9.9060e-03, -1.2510e-02,
40           -1.7415e-02,  2.9605e-02],
41           [ 1.7389e-02,  2.2764e-02,  1.1768e-03, -3.7927e-02,  9.6036e-03,
42           -1.1897e-02, -2.3105e-02],
43           [ 2.3286e-02, -1.9286e-02, -2.9396e-02,  7.3536e-04, -4.1443e-03,
44           3.5126e-03, -3.3107e-02],
45           [-1.4172e-02,  1.7171e-02, -1.8400e-02, -4.5913e-02, -1.2173e-02,
46           4.0021e-02,  9.6844e-03],
47           [-2.4347e-03, -2.3539e-02,  3.9338e-03,  1.4739e-02,  8.2878e-03,
48           -1.4087e-03, -5.7523e-03],
49           [-2.1811e-02,  1.6907e-02,  9.2821e-03, -5.7126e-02,  1.1346e-02,
50           4.3896e-02,  1.5167e-02]]]],
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54      [-1.1766e-02, -1.3360e-02, -1.8511e-05,  3.3169e-02, -2.4392e-02,
55          -3.1785e-02,  5.7032e-02],
56      [-1.5514e-02, -2.8294e-02, -8.3164e-03,  4.5450e-03,  3.2530e-02,
57          -4.2760e-02, -4.2544e-02],
58      [-2.8673e-02, -3.9962e-02, -2.0781e-02,  4.2311e-02,  3.5678e-03,
59          -1.4418e-02, -5.8129e-02],
60      [ 2.6300e-02, -2.3049e-02,  3.2509e-02, -1.8341e-03, -2.9896e-02,
61          4.0993e-02,  1.4169e-02],
62      [ 2.0684e-02, -2.1201e-02, -3.0531e-02,  8.6117e-03,  3.6190e-02,
63          -7.9796e-03, -4.1934e-02],
64      [ 2.8486e-02,  7.4033e-03, -2.8891e-02,  2.2850e-02,  4.1698e-03,
65          -3.1728e-02, -3.8636e-02],
66      [ 3.1227e-02, -2.9698e-02, -2.3472e-02, -4.5807e-03,  2.9264e-02,
67          1.2792e-02, -3.3334e-02],
68      [-7.3751e-03, -5.4200e-03,  2.2617e-02,  7.9912e-03, -1.3240e-02,
69          -5.5487e-03,  1.9709e-02],
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71          6.1295e-03, -3.6361e-02]],
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74          -4.5959e-02, -2.3326e-03],
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77      [ 1.0560e-02,  5.8066e-02, -1.5669e-02, -6.6895e-02, -7.5045e-03,
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79      [ 2.0318e-02,  7.3558e-02, -3.4476e-02, -4.6549e-02, -1.5188e-02,
80          -4.4780e-03, -1.1969e-02],
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85      [ 1.0504e-02,  2.3087e-02, -3.1595e-02, -3.0036e-02,  1.6276e-02,
86          -1.2534e-02, -1.4531e-02],
87      [-1.0467e-02,  2.8002e-02,  1.4892e-02, -2.1192e-03, -2.4797e-02,
88          -3.2195e-02, -2.8042e-02],
89      [-1.5378e-02, -7.7537e-02,  1.3504e-03,  3.5516e-02, -9.3421e-03,
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91      [-1.3743e-02,  5.3622e-02,  1.0131e-02, -5.0873e-02, -3.0842e-02,
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96      [-1.8156e-02, -3.7147e-02,  1.0373e-02, -3.3690e-03, -9.9667e-04,
97          -1.9709e-02,  2.2545e-02],
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99          -1.0230e-02, -2.6342e-02],
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131      [ 1.9298e-02,  1.5084e-03,  2.0689e-02, -1.7850e-02,  6.4230e-03,
132      3.3818e-02,  9.2802e-03],
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136      [[-5.7019e-03,  2.7035e-02,  1.5788e-02, -3.3354e-02,  1.4931e-02,
137      1.5869e-02, -4.0662e-02],
138      [ 5.2745e-03, -8.6815e-03,  1.8134e-02, -8.7430e-03, -3.9628e-02,
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142      [ 1.1108e-02, -1.8279e-02,  2.6924e-02,  1.0969e-02,  2.5283e-02,
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146      [ 2.2614e-02, -2.9893e-02, -1.4151e-02,  2.7910e-02,  3.0196e-02,
147      -2.4989e-03, -3.3879e-02],
148      [ 1.4620e-02, -3.5948e-02, -4.0909e-03,  9.3032e-03,  5.3325e-03,
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174      2.6250e-02, 6.3682e-03],
175      [-7.9176e-03, 5.5330e-03, 2.1059e-02, -8.8809e-03, -4.8668e-02,
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178      [[-2.4998e-02, 3.4489e-02, 1.4433e-02, -5.7222e-02, -7.5889e-03,
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182      [ 2.9809e-02, -7.0665e-02, -2.3570e-02, 5.0752e-02, 1.0169e-02,
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184      [ 2.5266e-02, -3.5426e-02, -5.1430e-03, 5.7812e-02, 2.6245e-02,
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235        2.5905e-02, -1.5949e-02],
236      [-1.9798e-02,  2.6335e-02,  1.5134e-02,  1.8301e-02, -1.8596e-03,
237        8.6250e-03, -9.5561e-03],
238      [ 1.5323e-02,  3.6910e-02, -6.0814e-03, -2.0757e-02, -2.3516e-02,
239        2.3859e-02,  2.7249e-02]]], device='cuda:0', requires_grad=True)
240 Parameter containing:
241 tensor([[[ 1.5856e-02,  2.1024e-02, -2.6273e-02, -5.5305e-03,  1.9890e-02,
242            1.2832e-02, -2.2681e-02]],
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244          [[ 2.9249e-02,  9.2978e-02, -3.4449e-02, -6.6374e-02,  4.1601e-02,
245            2.3132e-02, -3.0528e-02]],
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249
250          [[ 7.5089e-03,  3.0004e-02, -1.2281e-02, -1.8540e-02,  3.4121e-02,
251            3.8474e-02, -1.1338e-02]],
252
253          [[ 1.3426e-02,  1.2298e-02, -1.7902e-02,  3.4562e-02,  2.7771e-02,
254            -1.8773e-03, -4.6686e-02]],
255
256          [[ 8.3610e-05,  4.4821e-02, -2.6288e-02, -2.2594e-02,  2.0257e-02,
257            2.9546e-02,  3.1661e-03]],
258
259          [[ 2.5325e-02,  3.4688e-02,  5.3248e-03, -8.7550e-03,  2.1872e-02,
260            1.3269e-02,  3.9677e-02]],
261
262          [[ 8.9211e-03, -3.9193e-02, -2.2070e-02,  1.4060e-02,  2.5399e-02,
263            -4.4074e-02, -3.0133e-02]],

```

```

264
265     [[-1.5876e-02,  1.0415e-02, -2.0999e-02, -7.3541e-03,  3.6677e-02,
266        -6.0140e-03, -4.2536e-02]],
267
268     [[ 2.4456e-02, -5.6268e-02,  1.5242e-02,  3.4449e-02, -2.0776e-03,
269        -4.1596e-03,  1.7123e-02]]], device='cuda:0', requires_grad=True)
270 =====
271
272 =====
273 Layer (type:depth-idx)          Output Shape          Param #
274 =====
275 CPIKANModel                    [1, 1]                --
276 |-----ModuleList: 1-1        --                --
277 |   |-----0.weight            |-----70
278 |   |-----1.weight            |-----700
279 |   |-----2.weight            |-----70
280 |   |-----ChebyKANLayer: 2-1  [1, 10]              70
281 |   |   |-----weight          |-----70
282 |   |-----ChebyKANLayer: 2-2  [1, 10]             700
283 |   |   |-----weight          |-----700
284 |   |-----ChebyKANLayer: 2-3  [1, 1]              70
285 |   |   |-----weight          |-----70
286 =====
287
288 Total params: 840
289 Trainable params: 840
290 Non-trainable params: 0
291 Total mult-adds (Units.MEGABYTES): 0.00
292 =====
293
294 Input size (MB): 0.00
295 Forward/backward pass size (MB): 0.00
296 Params size (MB): 0.00
297 Estimated Total Size (MB): 0.00
298 =====
299
300
301 进程已结束, 退出代码为 0
302

```